

Flip-Flop Conversion

Make one type of flip-flop from another · the 4-step method · fast tricks · drawn circuits

QUICK TRICK

Learn the two fast tricks first (details below):

1. To make a D flip-flop: $D = \text{the next state you want}$. **2. To make a T flip-flop:** $T = Q \oplus (\text{the next state you want})$.

Most conversions in the exam can be done with these two ideas and a few simple gates.

12.1 What is flip-flop conversion?

● LEVEL 1 · BEGINNER

The problem. You have one kind of flip-flop (say a JK) but you need another kind (say a T). **The idea.** Put a **small logic circuit in front** of the flip-flop you have, so that it behaves like the one you need.

Analogy. A travel plug adapter. The wall socket (your flip-flop) stays the same. The adapter (the small logic) changes how the plug (the new inputs) fits.

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New inputs ----> [ Small logic circuit ] ----> Inputs of the flip-flop you HAVE ----> Q
                  ^                                     |
                  +----- Q (fed back) -----+
  
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The new inputs act like the flip-flop you **want**. The small logic is what we have to find.

12.2 The method (4 steps)

● LEVEL 2 · EXAM

- STEP 1 Write the TRUTH TABLE of the flip-flop you WANT (the new one):
columns = present state Q , new inputs, next state $Q+$.
- STEP 2 For each row, look at the change $Q \rightarrow Q+$.
Use the EXCITATION TABLE of the flip-flop you HAVE to write what its inputs must be.
- STEP 3 Simplify each of those inputs with a K-map (use the X's!).
The variables are the new inputs and Q .
- STEP 4 Draw: new inputs $\rightarrow$ logic $\rightarrow$ old flip-flop.

The excitation tables (from Chapter 11). X means “don't care”. Keep this table in front of you:

Q	Q+	SR: S R	JK: J K	D	T
0	0	0 X	0 X	0	0
0	1	1 0	1 X	1	1
1	0	0 1	X 1	0	1
1	1	X 0	X 0	1	0

12.3 Worked example: SR → JK

● LEVEL 2 • EXAM

Goal: use an SR flip-flop and make it work like a JK flip-flop. **Why is this nice?** A JK has no invalid case, and an SR does. Our small logic must never give $S = R = 1$.

Step 1 and 2: the JK truth table ($Q, J, K \rightarrow Q+$) and, in the last two columns, the S and R needed for that change (from the excitation table).

Q	J	K	Q+	S	R
0	0	0	0	0	X
0	0	1	0	0	X
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	1	X	0
1	0	1	0	0	1
1	1	0	1	X	0
1	1	1	0	0	1

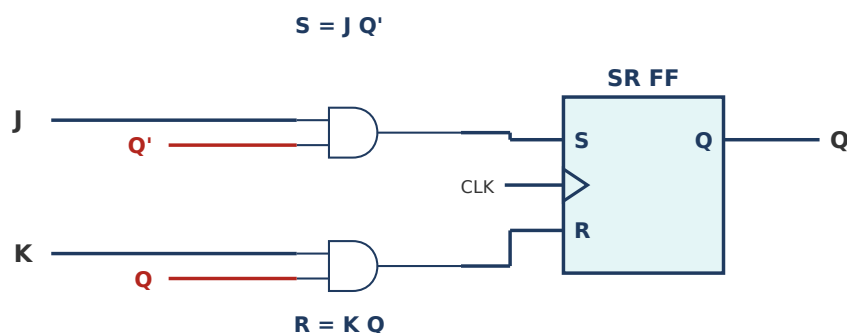
Step 3: K-maps (rows = Q, columns = J K). X cells can be used as 1 to make a bigger group.

		S			
Q \ JK		00	01	11	10
0		0	0	1	1
1		X	0	0	X

		R			
Q \ JK		00	01	11	10
0		X	X	0	0
1		0	1	1	0

S: the group m2, m3 ($Q = 0, J = 1, K$ any) gives $S = J Q'$. **R:** the group m5, m7 ($Q = 1, K = 1, J$ any) gives $R = K Q$.

Step 4: two AND gates in front of the SR flip-flop.



QUICK TRICK

Why is $S = R = 1$ impossible here? $S = JQ'$ needs $Q = 0$ and $R = KQ$ needs $Q = 1$. Q cannot be 0 and 1 at the same time. So the invalid case never happens. ✓

QUICK TRICK

Check with $J = K = 1$ (toggle): if $Q = 0$ then $S = 1, R = 0 \rightarrow$ set $\rightarrow Q = 1$. If $Q = 1$ then $S = 0, R = 1 \rightarrow$ reset $\rightarrow Q = 0$. It toggles. ✓

12.4 Fast tricks (skip the K-map)

● LEVEL 3 • GATE

Think about the **change you want**: the new flip-flop wants to go to a next state Q^+ (a formula in the new inputs and Q). Then:

Flip-flop you HAVE	Its input(s) must be	In words
D	$D = Q^+$	Just give it the next state you want.
T	$T = Q \oplus Q^+$	$T = 1$ only when the state has to change.
JK	$J = Q' \cdot Q^+ \quad K = Q \cdot (Q^+)'$	J: "go up" (from 0 to 1). K: "go down" (from 1 to 0).
SR	$S = Q' \cdot Q^+ \quad R = Q \cdot (Q^+)'$	Same as JK.

QUICK TRICK

Find Q^+ of the new flip-flop from its characteristic equation (Chapter 11):

SR: $Q^+ = S + R'Q$ | JK: $Q^+ = JQ' + K'Q$ | D: $Q^+ = D$ | T: $Q^+ = T \oplus Q$.

Then put it into the table above. That is the whole conversion!

Example: D → JK (you have D, you want JK)

New flip-flop is JK: $Q^+ = JQ' + K'Q$.

$D = Q^+ \rightarrow D = JQ' + K'Q$. (Two AND gates, one OR gate, one NOT for K' ; Q' comes from the flip-flop.)

Example: T → JK (you have T, you want JK)

$T = Q \oplus Q^+ = Q \oplus (JQ' + K'Q)$.

If $Q = 0$: $Q^+ = J$, so $T = J$. If $Q = 1$: $Q^+ = K'$, so $T = K$. Therefore $T = JQ' + KQ$.

Example: JK → D (the go-up / go-down trick)

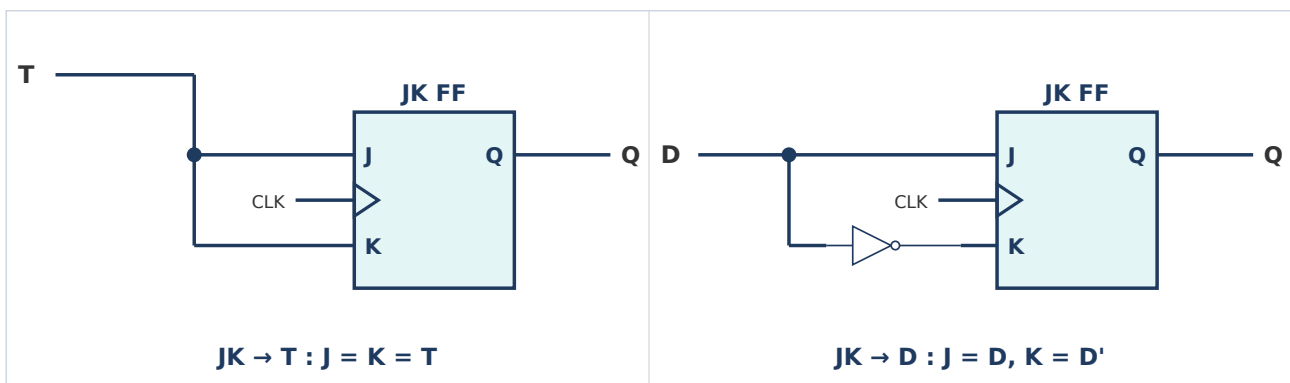
New flip-flop is D: $Q^+ = D$.

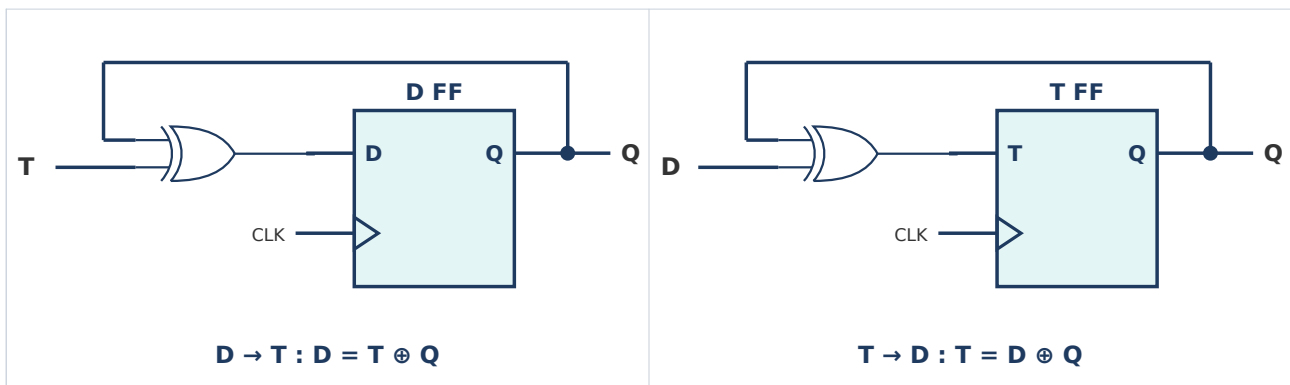
$J = Q' \cdot D$. $K = Q \cdot D'$. We can simplify: J can be D (when $Q = 1$ the J input does not matter, it is X) and K can be D' (when $Q = 0$ K does not matter). $J = D$, $K = D'$.

12.5 The four most common conversions (drawn)

● LEVEL 2 • EXAM

Here are the four conversions you are most likely to be asked about in the exam.





QUICK TRICK

JK → T needs no extra gate: just join J and K together and call the joint input T. (T = 0: J = K = 0 holds. T = 1: J = K = 1 toggles.)

JK → D needs one NOT gate (J = D, K = D').

D → T and T → D each need one XOR gate (and Q is fed back).

12.6 All conversions in one table

● LEVEL 2 · EXAM

Have ↓ Want →	SR	JK	D	T
SR	—	S = JQ', R = KQ	S = D, R = D'	S = TQ', R = TQ
JK	J = S, K = R	—	J = D, K = D'	J = K = T
D	D = S + R'Q	D = JQ' + K'Q	—	D = T ⊕ Q
T	T = SQ' + RQ	T = JQ' + KQ	T = D ⊕ Q	—

Extra gates needed:

Conversion	Extra gates
JK → T	None (join J and K)
JK → D	1 NOT
JK → SR	None (J = S, K = R)
D → T, T → D	1 XOR
SR → JK, SR → T	2 AND
SR → D	1 NOT
D → JK	2 AND + 1 OR + 1 NOT (for K')

GATE POINT

All twelve results in the table were checked by simulation: for every present state Q and every valid input, the converted circuit gives the same next state as the flip-flop it should imitate. (SR = 11 is excluded because it is invalid.)

12.7 Common mistakes

COMMON MISTAKE

- Using the characteristic table of the flip-flop you **have** instead of the one you **want**.
- Forgetting that the old flip-flop's inputs depend on **Q** as well (Q is fed back).
- Using the excitation table of the **new** flip-flop. You need the excitation table of the **old** one.
- Not using the X's (don't cares) in the K-map.
- Mixing up which flip-flop is “given” and which is “wanted”.

QUICK TRICK

Memory line: “Truth table of the **WANTED**, excitation table of the **HAVE**.”

Quick-Solve Tricks — Summary

Situation	Trick
Method	Truth table of the WANTED flip-flop + excitation table of the HAVE flip-flop + K-map.
Have D	$D = Q+$ (the next state formula of the wanted flip-flop).
Have T	$T = Q \oplus Q+$.
Have JK	$J = Q' \cdot Q+$ (go up), $K = Q \cdot (Q+)'$ (go down). Often $J = K = T$ or $J = D$, $K = D'$.
Have SR	$S = Q' \cdot Q+$, $R = Q \cdot (Q+)'$.
No extra gate	$JK \rightarrow T$ (join J, K) and $JK \rightarrow SR$ ($J = S$, $K = R$).
$D \leftrightarrow T$	Always one XOR with Q fed back.
$SR \rightarrow JK$	$S = JQ'$, $R = KQ$. The invalid state never happens.

GATE QUICK CHECK — Chapter 12

JK $\rightarrow$ T : $J = K = T$	JK $\rightarrow$ D : $J = D$, $K = D'$
D $\rightarrow$ T : $D = T \text{ xor } Q$	T $\rightarrow$ D : $T = D \text{ xor } Q$
SR $\rightarrow$ JK : $S = JQ'$, $R = KQ$	D $\rightarrow$ JK: $D = JQ' + K'Q$
T $\rightarrow$ JK : $T = JQ' + KQ$	SR $\rightarrow$ T : $S = TQ'$, $R = TQ$
SR $\rightarrow$ D : $S = D$, $R = D'$	

Traps: excitation table of the wrong flip-flop · not feeding Q back · forgetting that $JK \rightarrow T$ needs **no** gate · $D \leftrightarrow T$ needs an XOR (not an AND).

Practice questions

● LEVEL 1 · BEGINNER

Q1. How do you make a T flip-flop from a JK flip-flop?

Q2. How do you make a D flip-flop from a JK flip-flop?

Q3. How many extra gates are needed to make a T flip-flop from a D flip-flop?

● LEVEL 2 · EXAM

Q4. You have a T flip-flop and want a D flip-flop. Write T in terms of D and Q.

Q5. Write S and R for converting an SR flip-flop into a JK flip-flop.

Q6. Which gates are needed to make a JK flip-flop from a D flip-flop?

● LEVEL 3 · GATE

Q7. A T flip-flop is made from a D flip-flop with $D = T \oplus Q$. If T is held at 0, what does Q do?

- Q8.** An SR flip-flop is converted to a T flip-flop ($S = TQ'$, $R = TQ$). If $Q = 1$ and $T = 1$, find S , R and the next Q .
- Q9.** A T flip-flop is converted to a D flip-flop ($T = D \oplus Q$). If $Q = 1$ and $D = 0$, find T and the next Q .
- Q10.** A JK flip-flop is built from a D flip-flop ($D = JQ' + K'Q$). If $Q = 1$, $J = 1$, $K = 1$, find D and the next Q .
- Q11.** In the $SR \rightarrow JK$ conversion, why can $S = R = 1$ never happen?
- Q12.** Use the go-up/go-down trick: if you want the next state $Q^+ = T \oplus Q$ using a JK flip-flop, find J and K .

Answers

Q	Solution
1	Join J and K together and use the joint input as T ($J = K = T$).
2	J = D and K = D' (one NOT gate).
3	One XOR gate ($D = T \oplus Q$).
4	$T = Q \oplus Q^+$ and $Q^+ = D$, so $T = D \oplus Q$.
5	$S = JQ'$, $R = KQ$ (two AND gates).
6	$D = JQ' + K'Q \rightarrow$ two AND gates, one OR gate, and one NOT (for K'). Q' is taken from the flip-flop.
7	$D = 0 \oplus Q = Q$, so the flip-flop reloads its own value: Q holds (no change).
8	$S = 1 \cdot 0 = 0$, $R = 1 \cdot 1 = 1 \rightarrow$ reset $\rightarrow$ next $Q = 0$ (it toggled, as $T = 1$ should).
9	$T = 0 \oplus 1 = 1 \rightarrow$ toggle $\rightarrow$ next $Q = 0$ (equals $D = 0$ ✓).
10	$D = 1 \cdot 0 + 0 \cdot 1 = 0 \rightarrow$ next $Q = 0$ ($J = K = 1$ toggles $1 \rightarrow 0$ ✓).
11	$S = JQ'$ is 1 only when $Q = 0$. $R = KQ$ is 1 only when $Q = 1$. Both can never be 1 together, because Q cannot be 0 and 1 at once.
12	$J = Q' \cdot Q^+ = Q'(T \oplus Q)$. When $Q = 0$, $T \oplus Q = T$, so $J = T$. $K = Q \cdot (Q^+)' = Q \cdot (T \oplus Q)'$. When $Q = 1$, $T \oplus Q = T'$, so $(Q^+)' = T$ and $K = T$. Result: $J = K = T$ (this is the $JK \rightarrow T$ conversion).

Next: Chapter 13 — State Machines (state diagram, state table, Moore and Mealy machines, state reduction, sequence detector).